

## Part 1: Updated Progress Report 1

### 1. *Team Information*

Team 26: Media Map

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### 2. *Prior Individual Project*

We are expanding on Adi’s individual project. We reassessed the target user population by focusing on movies and TV shows exclusively (rather than books, movies, and TV shows), did more user research, changed the format entirely, and improved usability.

### 3. *Target User Population*

Our target user population is made of people in the teen to older adult age range who enjoy watching movies and TV shows and who want recommendations.

### 4. *Project Concept*

This project tackles a subproblem of a larger issue. The larger problem faced by users is effective media-finding: users want to get recommendations for film and television titles from algorithms, determine whether they want to watch it, and then access the shows.

Among other inefficiencies, the recommendation process is opaque. Users cannot understand, visualize, or control why a website’s recommendation algorithm is recommending a particular show, even if that would help them find the show they want. While other problems in the media-finding process are handled by other websites, like Netflix, IMDB, and Goodreads, there is no available source which allows users to clarify and control the recommendation process. Focusing on this subproblem lets our project serve an original and currently-unmet user need.

Our design solution is fundamentally based on letting the user decide on what criteria they would like to see recommendations. Our site’s back-end would group different film and television titles based on particular characteristics. Our site then uses these characteristics as a source of recommendations. The user can add a row to the site which filters movies by different criteria, like genre (e.g., ‘action’), style (e.g., ‘fast-paced’), or specific actors, directors, and other creators (e.g., ‘Alfred Hitchcock’). Within that row, users will receive movies and TV shows which fall under those criteria.

By allowing users to choose the specific, named connections which are causing movies and TV shows to be recommended to them, we are giving the users agency. We allow users to understand and control why they are being recommended a particular title.

### 5. *Challenge*

The project has a broad scope, and is a significant creative challenge. The problems users face—understanding and controlling their movie and television recommendations—is extremely broad. In order to give users the ability to control their own recommendation process, our design solution has to acknowledge many different priorities from users of all different kinds.

It also has to synthesize various different aspects of a movie or TV show in order to find similarities between different titles which exist for different reasons.

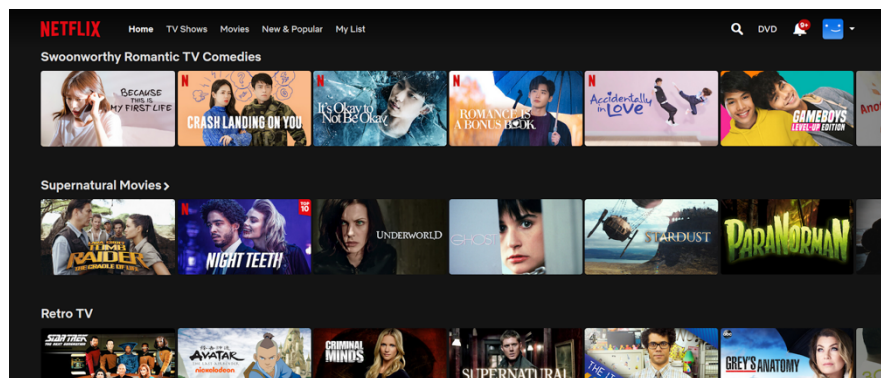
The project is also a creative design challenge. We are reimagining the traditional model of movie and TV recommendations in a way which must be innovative enough to engage users, intuitive enough to be usable, and which will give users understanding and control of the recommendation process. Figuring out how to let users best control their recommendations is a significant challenge.

## 6. Comparative Analysis

Many popular streaming services (Netflix, Hulu, Disney+, Peacock) use what we call the ‘Blockbusters’ model for recommendations. They show a horizontal array of movie and TV shows covers, similar to a shelf at the defunct video rental service Blockbuster, and each row is recommended based on a criterion chosen by the site, not the user.

These services do well by combining their central service—providing users with movies and TV that they can watch online—with recommendations. This creates a hub where users can find a show and immediately start watching it. They also do well by making the design similar to real-world designs—like shelves at a video store.

Where they fail is in their lack of control. Users cannot choose how movies and TV shows are recommended to them. Also, there is no way to get recommendations between streaming services—each service recommends only the titles that it owns, even if there are titles which the user would like better.



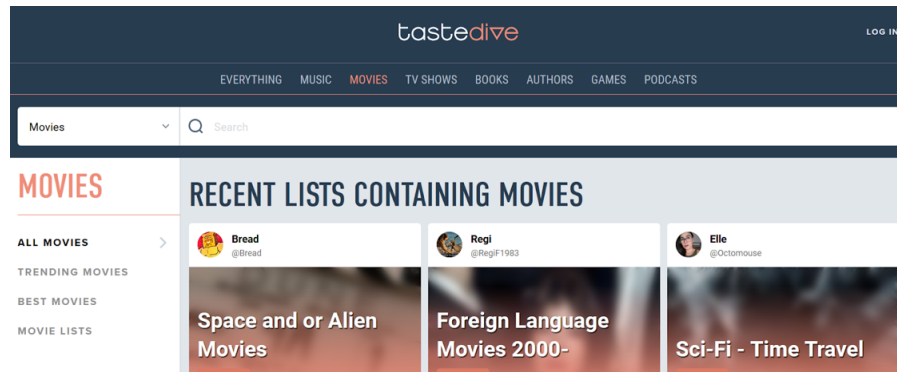
*A page from the popular streaming service Netflix showing horizontal rows of visual media. The rows are labeled ‘swoon worthy romantic TV comedies,’ ‘supernatural movies,’ and ‘retro TV.’ There is no way to choose what options are shown.*

There are also less-popular sites which are dedicated to finding new movies and TV shows, like TasteDive. TasteDive shows lists curated by individuals collecting different movies, TV shows, books, authors, games, podcasts, or music. It also shows which media titles of each type are ‘trending,’ and which are the ‘best.’ Clicking on an individual media title provides you with a brief summary of the story, links to streaming services where you can watch the title, and a number of media titles which are “similar” to it. No explanation is provided about how this similarity is determined.

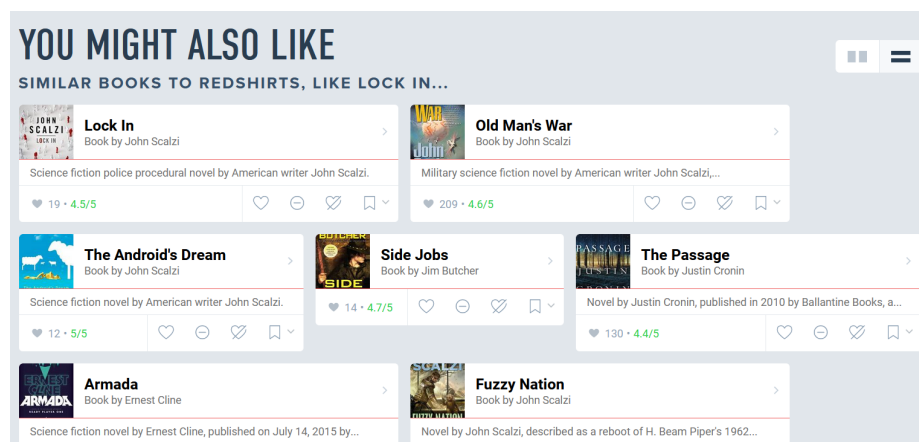
TasteDive succeeds by collecting many different types of media in one place. It allows users to see media titles from a number of streaming services, not just one, and gives users

access to them (by showing where users can stream that title). It also allows users to create lists and view the lists of other users. These are important efficiencies.

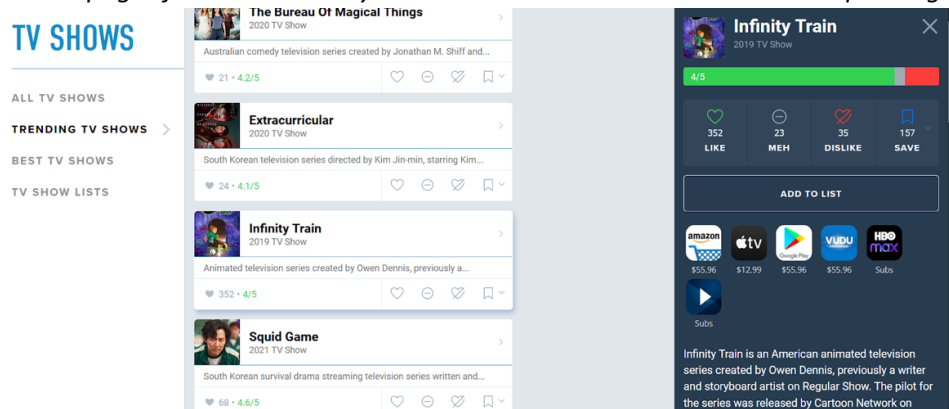
However, like the streaming services, TasteDive does not offer users full control of their recommendations. It also fails to explain how its recommendations are generated. The site's recommendations are based on the undefined characteristics of “trending,” “best,” or “similar,” or through the individually-curated lists.



*TasteDive's front page takes you to different types of media: 'everything,' music, movies, TV shows, books, authors, games, and podcasts. On the 'Movies' page, we see user-curated lists containing movies, trending movies, and 'best' movies.*



*Individual pages for a book show you books which are “similar” without explaining why.*



*TasteDive provides a summary of the media title, the ability to add it to a list, as well as links to where it is available to stream (and for how much money.)*

## 7. Explanation of Design Solution

As explained in our project concept, our goal is to provide more control on the user's end in terms of personalizing the recommendation algorithm of movies and TV shows. The basic premise of our design is that the user can add rows to the home page.

Our default screen starts with a pre-populated first row showing movies and TV shows which are currently popular. This row is a signifier: it demonstrates the affordances of the site by showing the user what a row will look like.

When designing the default screen, we kept several things in mind. Firstly, we used information design to emphasize the information we felt was most important for users. The most prominent part of the website is the website's title, 'Media Map,' and the search bar immediately under it. By making the title large and colorful, we help orient and center the users, and we draw attention to the search bar, which might otherwise go overlooked.

Next, the user's attention is drawn to the pre-populated first row, which has a number of colorful movie and TV show covers. For the rows, we decided to use a modular grid, with spacious gutters between each title. The gutters help keep the design minimalist and clean (as per Nielson's 8th heuristic, *aesthetic and minimalist design*.) The system of rows is inspired by other movie sites like Netflix (aligning with Nielson's 4th heuristic, *consistency and standards*) as well as physical movie distribution stores like the defunct Blockbusters (aligning with Nielson's 2nd heuristic, *match between system and the real world*.)

Finally, the user's attention goes to a colorful sidebar. The sidebar contains large, familiar icons, each one labelled with the function of its associated button.

Users can add their own rows to the site. If a user has specific criteria in mind, they can go to 'customize filters' on the sidebar. This will add two columns to the right of the sidebar. The central column lets the user choose which criteria (the *filters*) they want to include. The rightmost column lists the criteria which the user has chosen above an "add row" button. This column advances the user experience using Nielson's 5th and 6th heuristics: error prevention (the user is less likely to make a mistake if they can check what row they're creating before adding the row to the home page) and recognition rather than recall.

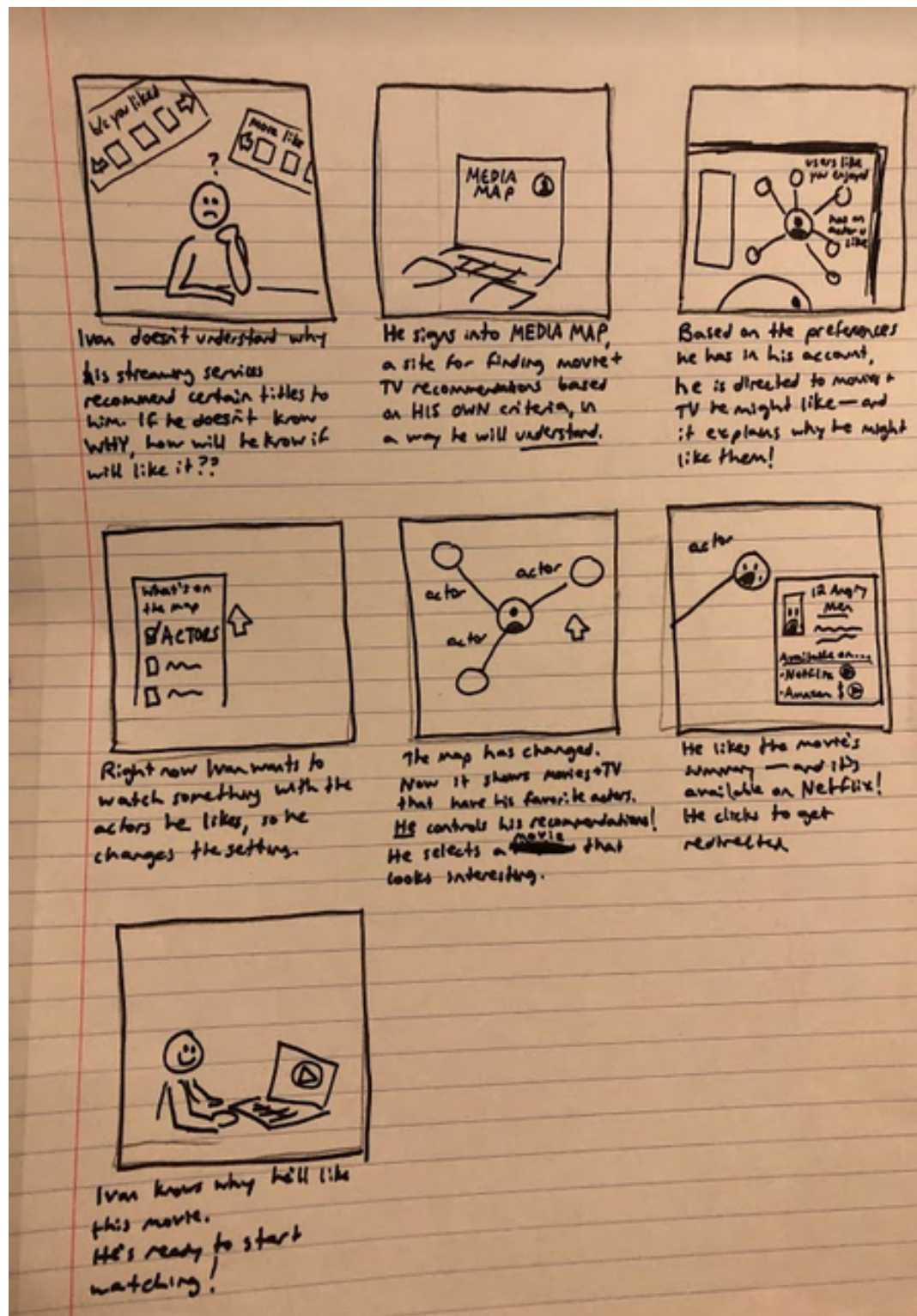
Users can also go to 'row library' to find both pre-made rows, and—if the user is signed in—the user's own saved rows. Pre-made rows may be especially convenient for novice users, while saved rows are particularly good for experienced users who do not want to recreate their own rows.

There are a number of other functionalities—collections, find friends, share—which are less relevant to our main design solution.

Finally, users can also search for more information about a specific movie. If the user clicks on one of the movies on the home screen, or searches for the movie, the user will see the movie's cover, a short summary, a list of streaming sites where that movie is available, and a list of the criteria which that movie falls under.

## 8. Storyboard(s)

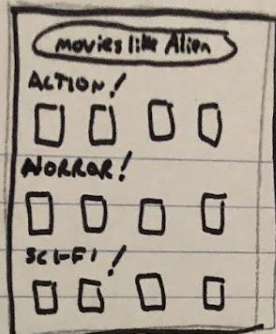
Note: these storyboards are based on an older format of our project. However, the user path has not changed—merely the design—so we felt that they remained applicable to our project.



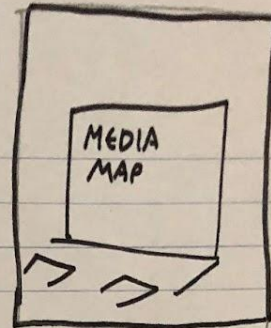




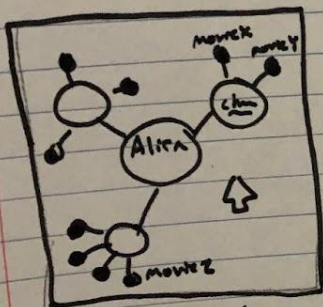
Nina is a big fan of the movie *Alien*, but only the characters — she normally hates that genre.



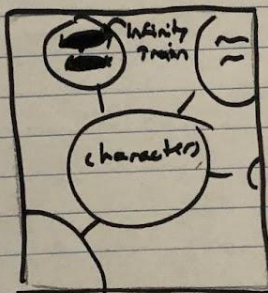
But traditional searches only give her movies + TV similar to *Alien*'s genre — not the characters! She can't control the recommendation to look for what she wants.



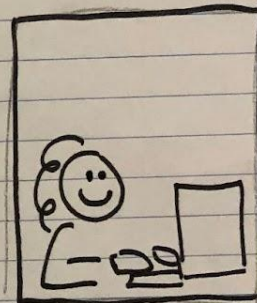
So she goes to Media Map, a site for controlling your recommendations.



Media Map shows her a map of other movies + TV shows with similar attributes to *Alien*.

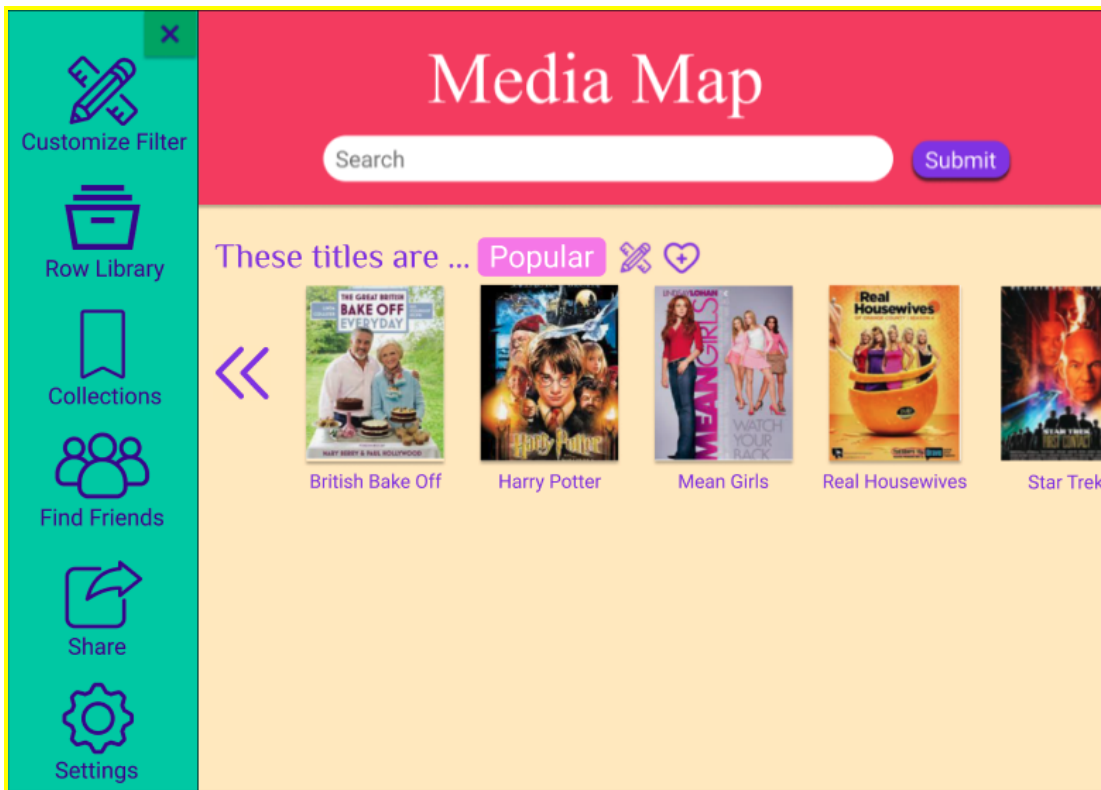
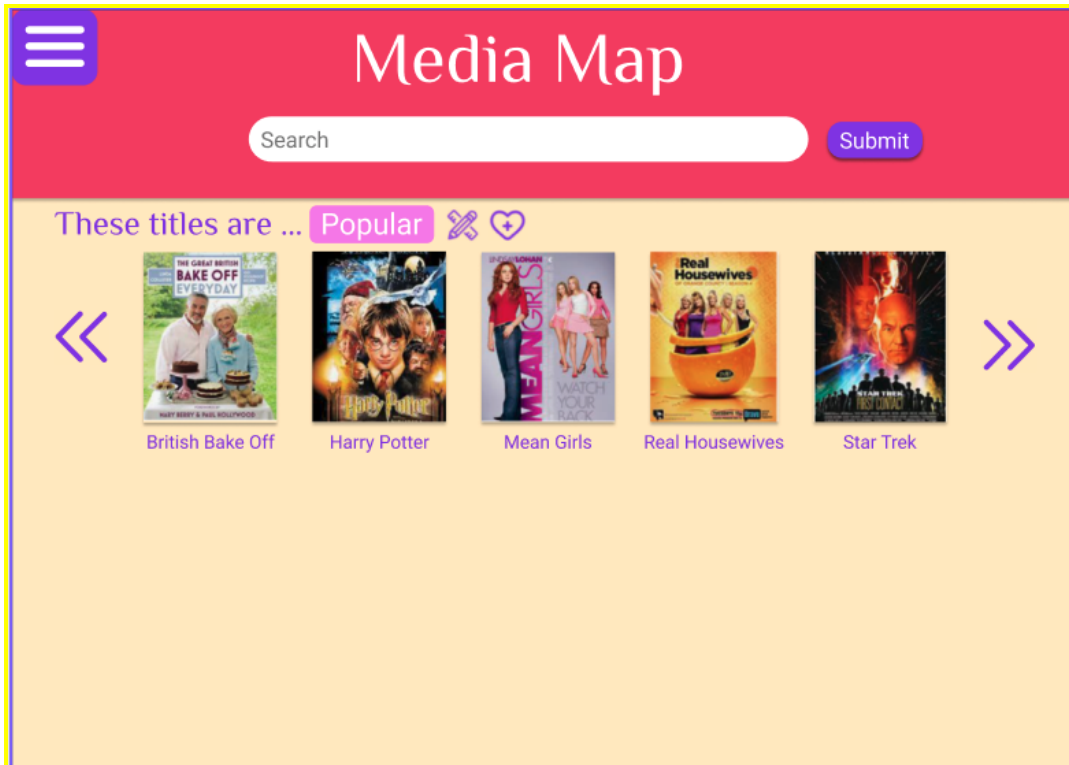


Look! Here are some movies + shows with similar characters, which aren't necessarily in the same genre!



Nina is happy. Thanks to Media Map, she got to a TV show that she wanted.

## 9. Current Prototype(s)



✕

Customize Filter

Row Library

Collections

Find Friends

Share

Settings

Customize Row

Genre

☐ Action
 ☐ Comedy
 ☐ Drama
 ☐ Horror
 ☐ Thriller
 ☐ Sci-Fi
 ☐ Mystery
 ☐ Fantasy
 ☐ Crime
 ☐ Romance

Style / Mood

☐ Feel-Good
 ☐ Serious
 ☐ Melancholy
 ☐ Intense
 ☐ Thoughtful
 ☐ Upbeat
 ☐ Mindless
 ☐ Goofy
 ☐ Thrilling
 ☐ Scary

Actor

Writer

Other Creator

Your Criteria

Add Row

Submit

Girls

Real Housewives

Star Trek

✕

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Row Library

Collections

Find Friends

Share

Settings

Customize Row

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 ☒ Comedy
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Comedy

Mystery

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Girls

Real Housewives

Star Trek





# Media Map

These titles are ... Popular  



British Bake Off



Harry Potter



Mean Girls



Real Housewives



Star Trek



These titles are ... Comedy  



Clue



Roger Rabbit



Knives Out



Detective  
Pikachu



iZombie



★ 7.3 / 10  
(Source IMDB)

## Clue: The Movie

Six blackmail victims are invited to an isolated mansion by a man who knows a dark secret from each of their pasts. On arrival, each is given a pseudonym drawn from Cluedo before being introduced to the blackmailer. Each is handed a weapon, at which point the lights are switched off and the blackmailer is killed. Can the guests uncover the murderer before they all become victims?

### Available On...

Paramount Vudu (\$) Microsoft (\$)  
Pluto (ads) iTunes (\$)  
Google Play (\$)

Click a criterion to add it to your current custom now. Click again to deselect.

Genre: Mystery  Comedy  Crime 

Style: Upbeat  Fast-paced 

Actors: Tim Curry  Madeline Khan  Christopher Lloyd  Eileen Brennan 

Writers: John Landis  Jonathan Lynn  Anthony E. Pratt(source material) 

Customize Filter  
 Row Library  
 Collections  
 Find Friends  
 Share  
 Settings

✕

Preset Rows

Saved Rows

Essential 80s

+

Popular movies from the 80s

Scarface
 Teen Wolf
 Predator

Mysteries

+

Works considered mysteries

Rebecca
 Da Vinci Code
 Castle

Horror Lovers

+

Spine-chilling horror films

Candyman
 Halloween
 Malignant

Media Map

Submit

Popular

✕

♡

Harry Potter
 Mean Girls
 Real Housewives
 Star Trek

Media Map

Submit

These titles are ...

Popular

✕

♡

«

British Bake Off

»

Harry Potter
 Mean Girls
 Real Housewives
 Star Trek

These titles are ...

Mysteries

✕

♡

«

Rebecca

»

The Da Vinci Code
 Castle
 Knives Out
 Midsomer Murders

## 10. Risks to Mitigate

At the moment, our main design risks are based on usability. We developed our final prototype based on a large number of user interviews. We were only able to take some of these responses into account due to time constraints. Further, our project would benefit from additional user testing across a diversity of demographics to ensure that the site is accessible, useful, and usable to a wide population.

Our project also has important technical risks. Will we be able to meet the standards set by the design? That is, generating the media map with simple and clear choices, such as “same actors” or “same director” is definitely feasible to implement. However, when if users choose some abstract criteria to generate the media map, such as “fast-paced,” it could be harder to provide better recommendations than existing recommendation services like Netflix. On top of that, after generating movie and TV show recommendations based on the user’s choice of key attributes, we would like to show users which streaming services they can utilize to watch each movie and TV show. However, as the streaming services update the list of movies and tv shows that are available based on regions, contracts, and so on, we might not be able to reflect these updates accurately. This means we which might end up providing outdated information.

## 11. Project Tasks

### - TASKS COMPLETED

- Pick design topic (2 hrs, all, 11/13)
- Balsamiq prototypes for different formats (2 hrs, Adi, 11/15)
- Further developed ideas for different formats (1 hr, Adi and Swan, 11/16)
- Conduct research on which Web Frameworks / DBMS we will use for the project (2 hrs, Je Seung and Aleksandra, 11/16)
- Create low-fi prototypes for different formats (2 hrs, Adi + Swan, 11/19)
- FP Progress Report I (all) 11/19
- First studio session (all) 11/23
- User interviews regarding different formats (all) 11/29
- New design based on user interviews (all) 12/01
- Lo-Fi prototype the new design (2hrs, Adi, 12/02)
- Coding planning & Research on technical stacks (3 hrs, Aleksandra and Je Seung, 12/02)
- FP Progress Report II and 360-assessment (all) 12/03
- Coding implementation (20 hrs, Aleksandra and Je Seung)
- Second studio session (15 min, all) 12/6
- User interviews regarding new design (30 min apiece, all) 12/9
- Update design based on secondary user interviews (30 min) 12/9
- High-fi prototype the updated design (10 hrs)
- Final project video (2 hrs)
- Final project turn-in (3 hrs)

## Part 2: Updated Progress Report 2

### 1. Design Process

See Final Progress Report, #4. Because there is a question for the Final Progress Report which also asks about our design process, we are including all loops of our design process there rather than repeat the same information.

## *2. Progress Report 1 follow-up*

### **● In our first meeting after Progress Report 1 feedback**

At our first final project studio, Professor Smith emphasized that the design portion of our project is more important than the technical implementation. In this regard, in our first post-studio team meeting, we decided to conduct user interviews with our five different Balsamiq prototypes, to fine-tune what design components we need to implement to make our project more user-friendly and useful. In that sense, we created interview questions and conducted sixteen user interviews over the week.

### **● In our second meeting after Progress Report 1 feedback**

In this meeting, we evaluated the result of user interviews, and then modified our design based on the user's response (as well as Professor Smith's feedback.) That is, since Professor Smith and most of the interviewees answered that the grid-style design is more usable due to its simplicity, we came up with the modified lo-fi design in the grid-style during the meeting. Then, we also modified smaller details on our final lo-fi design based on the user interview, to mitigate our design risks by finding the most optimal design.

## Part 3: Final Progress Report

### *1. Code Implementation Details*

We utilized the VScode editor to create the HTML/CSS files. We also used JQuery for the Javascript files to make the website dynamic and add stylistic properties to specific elements. For resources used, we used Github for version control, and W3Schools, Stack Overflow, and Mozilla Developer Web Docs to resolve issues and for element references.

### *2. Instructions to run your program*

- 1) Open site. You will see the default page. There is a pre-filled row. If you wish, you can use the arrows to go through the different movies on this row.
- 2) To open the sidebar, click the top left purple button. On the sidebar, select the icon labeled "row library."
- 3) This will open a side column which allows you to select either pre-made rows or saved rows. Click on the add icon next to the preset row, Mysteries. This will add a new row of mystery films and shows to your home page.
- 4) On the side bar, select the icon labeled 'customize filters.'
- 5) This will open two side columns. On the central column, there is a list of potential criteria for your new row. Select the criteria
  - (A) mystery
  - (B) comedy
  - (C) upbeat

*Note: due to the limits of our implementation, you can only add these criteria specifically, and in this order.*

- 6) Then select Add Row on the bottom of the selector screen. This will add a final third row to the home page. To exit out of the sidebar view, click on the X button on the top right of the sidebar. Scroll to the bottom of the home page and in the new row, find the movie *Clue* and select it.
- 7) This will open a modal page for the movie *Clue*. On the left side, there's the rating and movie summary. On the right-hand side, there is a box informing you of the streaming services that *Clue* is available on. Click on the Microsoft button and it will lead you to the Microsoft store page for *Clue*. To exit out of the modal, click on the X button on the top right corner.

### 3. *Progress Report 2 feedback follow-up*

Professor Smith suggested that we focus attention on the main purposes of the tools that we use as part of mitigating risk. Our tool's identity is formed by its *core value*: the value that it adds the user's life. He told us that we would get farther adding 5% to core value than 100% to a "side add." As such, we decided to focus our user interviews on what we could improve about our site's core value—letting users control their own recommendations—rather than other, less-important features, like finding friends, creating collections, or sharing content.

### 4. *Design Process*

#### **First Loop (informal)**

##### **- Biggest Risk**

Our biggest risk was deciding what problem to attack with our final project. We had to find a problem which really existed, but which we could solve in the time allotted.

##### **- Design Phase**

We used the 'design funnel' process in order to pick our design. Together, we brainstormed different ideas. We used a number of brainstorming strategies, including using a numbered list to identify different ideas, building off one another's ideas, and keeping it 'fun' (for example, by using lighthearted, goofy fonts on our shared Google Doc).

Ultimately, we decided that we wanted to continue Adi's Individual Project.

##### **- Evaluation Phase**

Because we were continuing an Individual Project, our needfinding step was already completed. Based on the results of the I.P., we determined that there was indeed a need for a Media Map that gave users control of their media recommendations. Moreover, the evidence from the user interviews of the I.P. suggested that there was a greater need for a Media Map restricted to only movies and TV shows.

However, based on the user interviews, the I.P. did not sufficiently prove that the proposed format was the best format for the project.



## **Second Loop**

### **- Biggest Risk**

The biggest risks we aimed to tackle with the UCD loop is to find the most optimal format for this project. Our project should allow users to visualize, understand, and control connections between different movies and TV shows, but there are many ways of doing this—some better than others. By creating numerous designs and showing them to users, we will climb different design hills until we find the format that is best for users: the “right design”.

### **- Design Phase**

In the second loop, extending from the first loop of the design process, we revisited the first step of the cycle, which is the design step. Based on the result of team meetings and the feedback we received on the Progress Report 1, we focused on re-evaluating our Need-finding and Storyboarding stage at first. As a result, we all agreed that there is no need to launch completely new need-finding or storyboarding steps. This is because our first user interviews conducted in the first design loop demonstrated that our intended target population, people who are interested in movies or TV shows, do indeed need and want our “Media Map.” They want to understand the logic behind why they are being recommended some titles, and control their recommendations.

However, although we did not need to do needfinding, we still had to deal with our biggest risk: how do we know what the optimal format is for our design? With this in mind, we created five different possible designs for Media Map. Three of these were various derivations off of a ‘bubble-style’ format, meaning that they displayed a visual map of different categories and media titles, each one enclosed in a circular “bubble.” Two others were grid-style, one with modular rows and another which displayed all the recommended movie and TV titles together as a single grid.

### **- Prototype Phase**

Once we had our designs, it was time to create Lo-Fi prototypes. Lo-Fi prototypes would allow us to could conduct user studies on the format of the design—thereby checking what formats and aspects users liked best, mitigating this loop’s biggest design risk—without wasting effort on a higher-fidelity prototype. So, we created five Lo-Fi prototypes in Balsamiq, include all three bubble-chart-style designs and two grid-style designs.

Based on Professor Smith’s feedback, we made the designs more individualized. Instead of having generic “movie name here” icons, we added the actual titles of movies ad TV shows, connected by actual links reflecting the content of those shows. Further, we had users provided us with the titles of shows that they liked in advance, and altered the designs further so that the design appeared personalized to the user’s tastes.

### **- Evaluation Phase**

After finishing those designs, we moved on to the third step of the design cycle, the evaluation step. As mentioned above, we have conducted user interviews (sixteen users in total) with the questions we created during our team meeting.

The list and structure of questions are like below:

#### PRE-INTERVIEW

1. How do you usually find movies and TV shows?
2. Can you tell me about the last X movies and TV shows you've watched? How did you find them?
3. What kind of things do you look for in a new movie or TV show?

*Show design 1.*

*Show design 2.*

*Show design 3.*

*Show design 4.*

*Show design 5.*

#### POST-INTERVIEW

1. What design was your favorite? Why? What did you like about it? What didn't you?
2. What would your ideal design be able to do?
3. If you could pick and choose different parts of different designs, what would you like?
4. What kind of "criteria" would you want to see on a website like this?

We constructed the interview structure like this, which is not to ask questions about each design before moving on to the next design, as we did not want to interrupt user's focus on exploring each design.

We switched the order of the designs we were showing. This means that some users saw bubble-style formats like Designs 1, 2, and 5 first, and other users saw grid-style formats like Designs 3 and 4 first. Due to constraints in synchronizing our work (since we all did interviews separately), as well as the constrained number of interviews done, the different orders of the designs were not truly randomly distributed. This is a potential issue in our research.

As a result of our interviews, we found that most of our interviewees preferred the grid style instead of the bubble-chart style, as it is simpler and easier to get the necessary information. This means the grid style appears to have a higher usability than the bubble-chart one. Users also enjoyed several aspects of the other designs, specifically the flexibility of the

5th (bubble-style) design. Therefore, as we will elaborate in the next design loop, we decided to move on with a renovated version of the grid-style design, which combined the grid format with additional functionalities that users liked.

### **Third Loop**

#### **- Biggest Risk**

In this loop, our biggest risk was *usability*. We now had an idea of what format our users liked best, but it was still vital to implement that format in an effective and usable way, so that users would be able to both understand and enjoy using our site. This means that our new design had to be clear, aesthetically-pleasing, and, if possible, follow design heuristics to improve the user experience.

#### **- Design Phase**

In order to account for our biggest risk—usability—we made sure to keep our user feedback from the evaluation phase of the prior loop closely in mind as we created a new design. By prioritizing user feedback, we would ensure that we were creating a product that our users would understand and enjoy: it would help us locate the “right design.”

We chose to use the grid-style format with modular rows, because this was the design that users found easiest to navigate and preferable to use. We also took other features of our interviews into account. In detail, we decided to organize our sidebar and top bars, since many of our interviewees expressed that they preferred a more organized filtering and search system similar to Design 5. Also, we decided to make the design more minimalistic, as many interviewees preferred that. On top of that, we have added the feature of share/export and friends/collections features. This is because some interviewees expressed their interest in sharing their media map filters to their friends as they do with Spotify (sending their Spotify playlist to their friends).

#### **- Prototype Phase**

After the design phase, we created a new prototype to reflect our new design: first a sketch outlining what we wanted our design to look like, then a more detailed, depth-first Balsamiq prototype. We chose to make the prototype depth-first—it demonstrates one specific user path in detail—in order to exemplify the most important workings of the website. While our site would theoretically have additional functionality that was not included in the prototype, we wanted to focus on improving the usability for the *core value* of our site. Our site is most useful because it helps users understand and control their movie and TV show recommendations: that is the value that it adds to a user’s life. Other tasks, like placing movies into collections or sharing things with friends, were less of a priority.

Although the prototype is more detailed, it is still a Lo-Fi prototype. We knew that we might have to change our design to make it more user-friendly, so a Lo-Fi prototype would prevent us from doing unnecessary work.

#### **- Evaluation Phase**

For the evaluation phase, we did over 13 user interviews with our depth-first Balsamiq prototype.

Overall, our users were happy with the usability of our product. They felt that the design was easy to use and very useful.

The most important feedback that our users had was regarding the *clarity* of our site. Our users had been confused by the labels “add custom row” and “add preset row.” They didn’t realize what a ‘preset row’ or ‘custom row’ was. They were also confused by the fact that the

pop-up column that appeared when they were editing row was labelled 'add custom row' even though they were not in fact adding a new row. Moreover, many interviewees disliked the two adjacent 'save row' and 'add/update row' buttons. They were confused by what 'saving' meant, and weren't sure what button to press, often choosing the wrong button at first.

Users also had ideas for improving the product's usability. Users also thought that the icons on the sidebar should be reordered—they wanted 'collections' to come first, then 'friends,' then 'share,' feeling that this was a more familiar ordering. Finally, users disliked the fact that the pop-up column which allows you to customize your filters (the column that appears after you click the 'add custom rows' button) forces you to scroll in order to see all the potential filter options.

#### **Fourth Loop**

##### **- Biggest Risk**

In this loop, our biggest risk was *improving usability, especially clarity*. Users had generally liked our design, so our baseline concept remained solid. However, we now knew where users were having trouble understanding and using our project. These trouble spots had to be fixed in order to improve the user experience.

##### **- Design Phase**

For our design phase, we again took user feedback from the previous loop's evaluation phase into account. This would help us mitigate our biggest risk: ensuring that the design was clear, understandable, and easy to use.

We changed the label 'add custom row' to 'customize filters.' The new label emphasizes what the user is actually doing: choosing which filters to use in order to get movie and TV show recommendations. Moreover, this name change avoids confusion when adding and editing rows. Both adding and editing rows occur in the same space: a pop-up column which allows the user to decide what filters will be used by a particular row, whether that row is being added or edited. We are now labelling this pop-up column based on the user's perception of it as a place to customize filters. This should improve users' ability to understand the purpose of this tool.

We also changed the process of saving rows. Now the 'customize filter' tool has only one button, 'Add Row,' which customizes the filters for a particular row. Saving rows now occurs on the home page, where users can see the results of a particular set of filters. The signifier of this affordance is a heart-shaped icon close to the row's label and edit button. This icon is similar to the 'save as favorite' button used by many other sites like Redbubble.

Below 'customize filters,' we changed the label 'add preset row' to 'row library.' We believe that "row library" will be a clearer name for users, since the concept of a library is familiar from the real world. The row library now includes both "pre-made rows" (a clearer term in this context than 'preset') and "saved rows."

Finally, we fixed the order of the icons on the sidebar. We also changed the spacing of the pop-up column which allows you to choose filters. This means that the user no longer has to scroll in order to see all the filters.

##### **- Prototype Phase**

For the prototype phase, we created a Hi-Fi prototype in both Figma and HTML/CSS. Once again, the prototype is depth-first. The greatest risk of this design loop is clarity and usability. We wish to improve the clarity and usability of the heart of our project—the recommendation process. In order to do that, it is most important to improve the full user path. Prototyping less-important parts of the site, like in a breadth-first search, is not a priority.

##### **- Evaluation Phase**

There is no evaluation phase for the fourth loop. We are turning the final project in at this point.

### 5. *Post-Mortems*

Adi:

I actually thought that our design process went very well. There were some minor bumps along the way—for example, I think the user interviews could have gone a lot better if our prototypes were more robust or if the intended user path was more clearly-defined, since my users would often try to use some aspect of the site that we hadn't added to the prototype. Overall, however, I think the process went very smoothly. If I were to do the project again, I might like to buttress our needfinding results. We used my individual project, and assumed that there was a user need for the Media Map based on my own interviews. I still believe that this is true—and all the users I interviewed for the final project agreed—but in the individual project, I interviewed a much smaller population of people, largely Barnard students, and I'm not certain that that was fully generalizable to a wider population. I think the project could have been aided by more quantitative and robust user testing.

Swan:

The design process went well since we iterated the user design loop repeatedly and made constant changes based on feedback from the users. If I were to do this project again, I would start from the layouts and interfaces that most users are already familiar with and tweak them into the service that we are trying to provide.

Je Seung (Jason):

Overall, I would like to say that our design process went well. Given that this project was starting off from Adi's Individual Project, we were able to focus on conducting the first loop of user interviews, which helped us a lot in terms of being able to modify the design quickly based on feedback. If I were to do this project again, I would like to interview more people from diverse backgrounds instead of narrowing down the interviewees' population to mostly our friends on campus due to the time constraints we have. By doing this, I think that this could give us a chance to come up with a more user-friendly design for a larger population.

Alexsandra:

Reflecting back, during the design process, it went well. The interviewees offered great feedback to help our team further refine our ideas and make essential decisions for our design, for example like the web versus grid layout of the map. If we were to complete this project again, next time we should interview more people of various backgrounds, especially in technical experience, using more fully flushed out prototypes so users can have an easier time visualizing the app concept and we can record their interactions with the app better.

### 6. *Optional permission to feature your project in Prof. Smith's teaching*

Adi, Alexsandra, Je Seung (Jason), and Swan give Prof. Brian A. Smith permission to feature this project in his teaching materials. Prof. Smith may refer to Adi, Alexsandra, Je Seung (Jason), and Swan by name when describing the project.